Simulations of Trapped Bose Gases with multiple Variational Monte Carlo methods

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Abstract

In this report, we simulate a Bose–Einstein condensate using two different variational Monte Carlo methods. We have implemented the Metropolis–Hastings algorithm and the standard Metropolis method. Furthermore, we have tested the robustness of our code by comparing the results to exact analytical values. In addition, we have compared the computational time required to evaluate the energy using analytical expressions and numerical derivatives. Moreover, we have computed column densities for a Bose–Einstein condensate in an elliptical trap and found that including interactions between the atoms significantly alters their spatial distribution and increases the spatial volume of the condensate.

Introduction

Bose-Einstein condensates (BECs) were theoretically predicted by Satyendra Nath Bose and Albert Einstein over a century ago. A BEC occurs when a gas is cooled to extremely low temperatures such that the a majority of the bosonic atoms occupies the ground state. In 1995, this state of matter was experimentally realized. This was achieved by cooling ^{87}Rb atoms to approximately 20 nK. The method used is known as evaporative cooling. This method confines a monatomic gas in a time-orbiting potential trap and releases the faster, higher-energy atoms [1] such that only the colder atoms remains trapped. Today, the production of BECs has become routine in laboratories and is a large research field with numerous applications. One of these applications is the use of BECs to complete precision measurements of gravity [2]. Another important application is the study of many-body quantum systems of bosons. A common physical description of BECs is given by the Gross-Pitaevskii (GP) equation. One of the assumptions required for this equation to be valid is that the gas has to be dilute. In this context, dilute means that the average distance between individual atoms is large, such that the interactions between them are weak. Today, scientists can use Feshbach resonances to increase the interaction strength. This can drive the system to a regime where the Gross-Pitaevskii approximation is no longer valid. To address this problem, we use numerical tools to model the BECs. In this project, we simulate a BEC with two different Variational Monte Carlo (VMC) methods. We simulate the trapped BEC with, and without inter-particle interactions and for different numbers of particles. In this report we first describe the theoretical framework of our confined BEC and the VMC methods used. We then present and discuss our results, and conclude with a summary of our findings.

Theory

Most BEC systems are described using the Gross–Pitaevskii (GP) equation, which is a mean-field approximation. The system is dilute and therefore satisfies $na^3 \ll 1$, where n is the particle density and a is the s-wave scattering length. In this regime, two-body interactions dominate and the scattering length fully characterizes the interaction. Within the GP framework, all particles are assumed to occupy the same condensate wavefunction, such that interactions are replaced by an average interaction field. The interaction strength can be tuned by varying the scattering length a. For sufficiently high a, the mean-field approximation breaks down and other methods must be used. To model this regime of interaction strength, we employ a Variational Monte Carlo (VMC) method, enabling an explicit treatment for the inter-particle correlations [3].

The physics

The Hamiltonian for the system is

$$H = \sum_{i}^{N} \left(-\frac{\hbar^2}{2m} \nabla_i^2 + V_{\text{ext}}(\mathbf{r}_i) \right) + \sum_{i<j}^{N} V_{\text{int}}(\mathbf{r}_i, \mathbf{r}_j), \quad (1)$$

where the index i runs over all N particles in the system. Constants such as $\hbar$ and m is set to one in our project for convenience. The external potential V_{ext} in Equation(1) is given by

$$V_{\text{ext}}(\mathbf{r_i}) = \begin{cases} \frac{1}{2} m \omega_{ho}^2 r_i^2 & (S) \\ \frac{1}{2} m \left[\omega_{ho}^2 (x_i^2 + y_i^2) + \omega_z^2 z_i^2 \right] & (E), \end{cases} \quad (2)$$

where (S) and (E) denote spherical and elliptical potentials, respectively. x_i, y_i and z_i represent the components of $\mathbf{r_i} = (x_i, y_i, z_i)$, while $r_i = |\mathbf{r_i}|$ is the distance between the atom and the center of the trap.

ω_{ho} represents the harmonic oscillator trapping frequency in the radial direction, while ω_z is the trapping frequency in the z-direction. The interaction between bosons is represented by a pairwise hard-core potential

$$V_{\text{int}}(r_{ij}) = \begin{cases} \infty & r_{ij} \leq a \\ 0 & r_{ij} > a, \end{cases} \tag{3}$$

where a is the scattering length of the particles and characterizes the interaction strength, while $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the distance between two particles. By using harmonic oscillator units

$$a_{ho} = \sqrt{\frac{\hbar}{m\omega_{ho}}}, \tag{4}$$

and energy in units of $\hbar\omega_{ho}$. We define dimensionless coordinates as

$$\mathbf{r}_i \to \frac{\mathbf{r}_i}{a_{ho}}. \tag{5}$$

With the use of the dimensionless coordinates we rewrite Equation (1) as

$$H = \sum_{i=1}^{N} \frac{1}{2}(-\nabla_i^2 + x_i^2 + y_i^2 + \gamma^2 z_i^2) + \sum_{i<j} V_{\text{int}}(|\mathbf{r}_i - \mathbf{r}_j|), \tag{6}$$

where $\gamma = \frac{\omega_z}{\omega_{ho}}$ is the anisotropy parameter of the trap. For a hard-core interaction, the potential is given by

$$V_{\text{int}}(r_{ij}) = \begin{cases} \infty & r_{ij} \leq a \\ 0 & r_{ij} > a, \end{cases} \tag{7}$$

where a is now expressed in units of a_{ho}. The parameter a is the s-wave scattering length, which characterizes low-energy two-body interactions in dilute Bose gases, see [4] for details.

The wavefunction

The variational trial wave function for the ground state of the system is

$$\Psi_T(\mathbf{R}, \alpha, \beta) = \left[\prod_i g(\alpha, \beta, \mathbf{r}_i)\right] \left[\prod_{j<k} f(a, |\mathbf{r}_j - \mathbf{r}_k|)\right], \tag{8}$$

where $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ represents the full configuration of the system, α and β are variational parameters of the wavefunction. The wavefunction in Equation (8) is described by a gaussian single-particle term $[\prod_i g(\alpha, \beta, \mathbf{r}_i)]$ and a two-body interaction term $\left[\prod_{j<k} f(a, |\mathbf{r}_j - \mathbf{r}_k|)\right]$ which is often named the Jastrow term. The gaussian single-particle term is a product of many standard harmonic oscillator ground state wavefunctions, which is defined as

$$g(\alpha, \beta, \mathbf{r}_i) = \exp\left[-\alpha(x_i^2 + y_i^2 + \beta z_i^2)\right]. \tag{9}$$

The two-body Jastrow term is given by

$$f(r_{ij}) = \begin{cases} 0, & r_{ij} \leq a, \\ 1 - \dfrac{a}{r_{ij}}, & r_{ij} > a, \end{cases} \tag{10}$$

and models interactions between atoms and enforces the hard-core repulsion. In the rest of the report we define $\Psi_T(\mathbf{R}) = \Psi_T(\mathbf{R}, \alpha, \beta)$.

Local Energy

The local energy is defined as

$$E_L(\mathbf{R}) = \frac{1}{\Psi_T(\mathbf{R})} H \Psi_T(\mathbf{R}). \tag{11}$$

where the kinetic term of Equation (11) is

$$\frac{1}{\Psi_T(\mathbf{R})} \sum_i^N \nabla_i^2 \Psi_T(\mathbf{R}). \tag{12}$$

Since we are doing numerical calculations we want numerical efficiency and stability. We therefore use the following identity for the trial wavefunctions when simulating the BEC

$$\frac{\nabla_i^2 \Psi}{\Psi} = \nabla_i^2 \ln \Psi + (\nabla_i \ln \Psi)^2.$$

With this we rewrite local energy into Equation (32) which you can find in Section Appendix. The analytical expression for the local energy without interactions is as in Equation (33). The analytical expression for the local energy, with interactions is given in Equation (34). For N non-interacting bosons, the exact total ground-state energy is

$$E_{\text{exact}} = N\frac{\hbar}{2}(2\omega_{ho} + \omega_z). \tag{13}$$

The probability distribution of the particles positions is given by the Born rule and is

$$P(\mathbf{R}) = \frac{|\Psi_T(\mathbf{R})|^2}{\int |\Psi_T(\mathbf{R})|^2 d\mathbf{R}}. \tag{14}$$

Variational Monte Carlo Methods

The theory of Monte Carlo methods, optimization and the blocking method is taken from [5].

VMC for the system of BEC

VMC is based on the variational principle from quantum mechanics. The variational principle states that the expectation value of the Hamiltonian with respect to any trial wavefunction Ψ_T provides an upper bound to the true ground-state energy E_0

$$\langle H \rangle = \frac{\int d\mathbf{R}\, \Psi_T^*(\mathbf{R})\, H(\mathbf{R})\, \Psi_T(\mathbf{R})}{\int d\mathbf{R}\, \Psi_T^*(\mathbf{R})\, \Psi_T(\mathbf{R})} \geq E_0. \tag{15}$$

By optimizing the variational parameters in the trial wavefunction, one can minimize the expectation value of the energy and obtain the best approximation to the true ground state. The trial wavefunction we use for this project is given in Equation (8). The expectation value of the local energy for a given wavefunction is

$$\langle E_L(\mathbf{R}) \rangle = \langle H \rangle = \int P(\mathbf{R})\, E_L(\mathbf{R})\, d\mathbf{R}, \tag{16}$$

where $P(\mathbf{R})$ is defined in Equation (14) and $E_L(\mathbf{R})$ is the local energy for configuration $\mathbf{R}$. The discrete expression of the expectation value of the local energy that the simulation will calculate is

$$\langle E_L(\mathbf{R})\rangle \approx \frac{1}{M}\sum_{i=1}^{M} E_L(\mathbf{R_i}), \tag{17}$$

where $E_L(\mathbf{R_i})$ is the energy evaluated for a specific configuration $\mathbf{R_i}$ and M is the total number of Monte Carlo cycles. The explicit expressions for $E_L(\mathbf{R_i})$ is given in Equation (34) and (33), corresponding to the case with and without interactions, respectively.

Brute-force Metropolis

The brute-force Metropolis algorithm is an algorithm that stochastically draws samples from a probability distribution and uses these samples to estimate expectation values. The specific method used for this project is explained in the algorithm below. The acceptance rule is

$$A(\mathbf{R}\to\mathbf{R}') = \min\left(1, \frac{P(\mathbf{R}')}{P(\mathbf{R})}\right),$$

where $P(\mathbf{R})$ and $P(\mathbf{R}')$ are the probabilities of the system being in configurations $\mathbf{R}$ and $\mathbf{R}'$, respectively. We use this acceptance rule in logarithmic form in our algorithm for numerical stability

$$\log\lambda = \log P(\mathbf{R}') - \log P(\mathbf{R}). \tag{18}$$

The brute-force metropolis algorithm is

Algorithm 1 Brute-force Metropolis

1. Define the variational parameter α, the number of Monte Carlo cycles M, and an initial configuration $\mathbf{R} = \{\mathbf{r}_1, \ldots, \mathbf{r}_N\}$
2. For each Monte Carlo cycle do:
   (a) Choose a random particle index $i \in \{1, \ldots, N\}$
   (b) Propose a new position:
   $$\mathbf{r}_i' = \mathbf{r}_i + S\eta, \qquad \eta \sim \mathcal{N}(0,\sigma^2), \qquad S = \text{scale}$$
   (c) Construct the proposed configuration $\mathbf{R}'$
   (d) Compute
   $$\log\lambda = \log P(\mathbf{R}') - \log P(\mathbf{R})$$
   (e) Draw $u \sim \mathcal{U}(0,1)$
   (f) Accept the move if
   $$\log u < \min(0, \log\lambda)$$
   (g) If accepted, set $\mathbf{R}\to\mathbf{R}'$; otherwise keep $\mathbf{R}$
   (h) Evaluate the local energy for the current configuration and store it
3. After all Monte Carlo cycles, compute the mean local energy

The scale in step $2(b)$. is chosen such that you get an acceptance rate of roughly $0.6 - 0.85$. The algorithm above was for one specific alpha and usually there is another loop outside, looping through different variational parameters.

Importance Sampling and Metropolis Hastings with Langevin drift

The reason to implement importance sampling is to make the sampling more effective. Importance sampling pushes the random walker towards regions with a higher probability density. The importance sampling we are going to use is derived from the Fokker Plank equation

$$\frac{\partial P}{\partial t} = D\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x} - F\right)P(x,t) \tag{19}$$

and the corresponding Langevin equation

$$\frac{\partial x(t)}{\partial t} = DF(x(t)) + \eta, \tag{20}$$

where F is the drift term and η is a Gaussian random variable. We do not derive these equations here, and refer to [5] for details. Following from the equations above, we get that the proposal position for particle i is given by

$$\mathbf{r}_i' = \mathbf{r}_i + D\mathbf{F}_i(\mathbf{R})\Delta t + \sqrt{2D\Delta t}\,\eta, \tag{21}$$

where $\eta \sim \mathcal{N}(0,1)$, D is the diffusion coefficient, which we set to $D = 0.5$, and Δt is a chosen time step which you can change for optimization. The Langevin drift force $\mathbf{F}_i(\mathbf{R})$ in Equation (21) is defined as

$$\mathbf{F}_i(\mathbf{R}) = \frac{2\nabla_i \Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})} = 2\nabla_i \ln \Psi_T(\mathbf{R}). \tag{22}$$

The analytical expression for the quantum force for a system with and without interactions are in Equation (38) and (39), respectively. The acceptance probability is given by the Metropolis Hastings algorithm,

$$A(\mathbf{R} \to \mathbf{R}') = \min(1, q(\mathbf{R}', \mathbf{R})), \tag{23}$$

where

$$q(\mathbf{R}', \mathbf{R}) = \frac{G_i(\mathbf{R}, \mathbf{R}', \Delta t)\, P(\mathbf{R}')}{G_i(\mathbf{R}', \mathbf{R}, \Delta t)\, P(\mathbf{R})}. \tag{24}$$

Here, $G_i = G_i(\mathbf{R}', \mathbf{R}, \Delta t)$ is the Green's function for the Langevin process and is defined as

$$G_i = \frac{1}{(4\pi D \Delta t)^{d/2}} \exp\left(-\frac{(\mathbf{r}_i' - \mathbf{r}_i - D\mathbf{F}_i(\mathbf{R})\Delta t)^2}{4D\Delta t} \right), \tag{25}$$

where d represents spatial dimensions of the system. For our project, the acceptance ratio is computed in log form

$$\log \lambda = \log \frac{P(\mathbf{R}')}{P(\mathbf{R})} + \log G(\mathbf{R}, \mathbf{R}', \Delta t) - \log G(\mathbf{R}', \mathbf{R}, \Delta t), \tag{26}$$

which improves numerical stability. The algorithm is as follows

Algorithm 2 Metropolis–Hastings with Langevin drift

1. Define the variational parameter α, the number of Monte Carlo cycles M, and an initial configuration $\mathbf{R} = \{\mathbf{r}_1, \ldots, \mathbf{r}_N\}$
2. For each Monte Carlo cycle do:
   (a) From the configuration $\mathbf{R}$, choose a random particle index $i \in \{1, \ldots, N\}$
   (b) Compute drift force $\mathbf{F_i}(\mathbf{R})$ and pick $\eta \sim \mathcal{N}(0,1)$
   (c) Propose a new position:
   $$\mathbf{r_i'} = \mathbf{r}_i + D\mathbf{F_i}(\mathbf{R})\Delta t + \sqrt{2D\Delta t}\,\eta,$$
   where $\Delta t = scale$
   (d) Construct the proposed configuration $\mathbf{R}'$
   (e) Compute $\mathbf{F_i}(\mathbf{R}')$ with Equation (22) and log-probability $\log \lambda$ with Equation (26)
   (f) Draw $u \sim \mathcal{U}(0,1)$
   (g) Accept the move if
   $$\log u < \min(0, \log \lambda)$$
   (h) If accepted, set $\mathbf{R} \to \mathbf{R}'$; otherwise keep $\mathbf{R}$
   (i) Evaluate the local energy for the current configuration and store it
3. After all Monte Carlo cycles, compute the mean local energy

Optimization Methods

Optimization methods are often implemented to minimize computational expense such that finding the optimal variational parameter takes less time. For our project we want to implement the method called steepest descent. This method updates the variational parameter in the direction of where the energy decreases. As a result, the program evaluates fewer energies for values of alpha far from the one that correspond to the true minimum of the energy. In practice you start by initializing your training by picking one starting alpha, α_0. The variational parameter is then computed with

$$\alpha_{k+1} = \alpha_k - \gamma_k \frac{d\langle E_L(\alpha_k)\rangle}{d\alpha} \tag{27}$$

where γ_k is the learning rate and can be updated or held constant for each iteration while $\frac{d\langle E_L(\alpha_k)\rangle}{d\alpha}$ is the gradient of the energy with respect to α. For our project we choose to hold the learning rate constant as in the standard steepest descent. For our particular system, we know that the $E(\alpha)$ function is convex, which makes steepest descent suitable as a method. The reason for this is that a convex function only has one minimum, meaning that the algorithm does not accidentally fall into a local minima during training.

For more about steepest descent and convex functions read chapter 6 in [5].

One limitation of steepest descent is that it requires an initial guess of the variational parameter. If you guess an alpha that is far from the optimal value, it can take a long time for your model to converge if you have a low learning rate. This is because a low learning rate corresponds to small jumps in the parameter you try to optimize. On the other hand, if you increase the learning rate too much, you can end up with unstable updates. This means that the algorithm overshoot the minimum and you end up failing to find the optimal value.

When running your VMC algorithms it is crucial that the program stop. You can then either program it to stop at a finite number of iterations, or you can implement a tolerance and patience. For each update of alpha you calculate the difference between the current estimated energy and the lowest one $\left|E_L^{min} - E_L^i\right|$. If this difference does not exceed the tolerance for n iterations in a row, the program stops. This n is what we call the patience.

Statistical analysis with the blocking method

To obtain a reliable estimate of the variance and standard deviation (std) for a VMC estimate, a statistical analysis of the samples is needed. This is because the consecutive samples in a Monte Carlo chain are correlated, which will affect the variance and std. To counter this you can use a method called blocking. Blocking is a method where you divide your samples into different blocks and find the block average of the variable you are sampling

$$\bar{E}_k = \frac{1}{B}\sum_{i=1}^{B} E_i, \tag{28}$$

here B is the number of samples for this block and E_i is the i-th sample. These block averages are approximately uncorrelated with each other which means you can use the standard equation for the variance and standard error of the mean. The variance is then

$$\mathrm{Var} = \frac{1}{n-1}\sum_{k=1}^{n}\left(\bar{E}_k - \bar{E}\right)^2, \tag{29}$$

where $\bar{E}$ is the mean of the block averages and n is the number of blocks. The standard error of the mean is then given by

$$\sigma_{\bar{E}} = \sqrt{\frac{\mathrm{Var}}{n}}, \tag{30}$$

which provides an estimate of the statistical uncertainty in the VMC energy. As the block size increases the correlation between samples decreases. Therefore, it is standard practice to plot the standard error of the mean as a function of block size and observe where convergence occurs. This is also important since, for very large block sizes, there are too few samples, leading to increased statistical noise.

Column Density

Column density is obtained by integrating out one dimension of the Onebody density

$$\rho_{\mathrm{col}}(x,y) = \int \rho(x,y,z)\,dz, \tag{31}$$

where $\rho(x,y,z)$ is the Onebody density. $\rho(x,y,z)$ is the probability density of finding a particle in a given volume in space. For a harmonic oscillator with an elliptical potential in three dimensions, it is more informative to study the column density due to the cylindrical symmetry. For the discrete case the column density is found by first defining a suitable number of annuli with different fixed radial distances in the x, y-plane. For each MC cycle, count the number of particles that are located within each annulus. This creates a histogram where the bins represent distance from the origin of the trap. After the simulation, the number of particles in each bin is divided by its corresponding annular area and by the number of MC cycles. This yields a discrete representation of the column density.

Implementation

The implementation is divided into different steps. We first validate the method in the non-interacting case, followed by improvements in the sampling efficiency, parameter optimization, and uncertainty estimation. Lastly, we implement the interacting case and different methods to compare the system with and without interactions.

Our goal is to create a program that can simulate a BEC for different variational parameters in the wavefunction, both with and without interactions and for different numbers of atoms and dimensions. The methodology for this project is to first implement a standard Metropolis algorithm that calculates the local energy for a given alpha, without interactions, and then expand on this once the code is verified. The expression for the exact ground state energy and the optimal variational parameter is known, and we can therefore directly compare our simulation results with the exact values. With this approach we can test wether the simulation yields correct results after implementing a new numerical tool.

We first implement the standard Metropolis algorithm to calculate the local energy for a given alpha. We calculate the VMC energy for different values of α and plot energy vs alpha. From this plot we locate the minimum. The value of α corresponding to the minimum is the optimal variational parameter. We do this for dimensions, $d \in \{1, 2, 3\}$, and number of particles, $N \in \{1, 10, 100, 500\}$, and compare with the exact ground state energy. After confirming that our model computes the energy correctly, we run tests to compare computational time per Monte Carlo cycle for analytical and numerical evaluation of the energy.

After this, we implement importance sampling using the Fokker-Planck and Langevin equations. The

reason we do this is that the drift term guides the walker towards a region with higher probability density. This increases the efficiency of the sampling compared to the standard brute-force metropolis. We then test the new algorithm for sampling and compare to the results we got for the exact solution earlier. We also test different step sizes dt to determine which value of dt that is optimal for sampling.

Since a coarse scan over many values of α requires a large number of MC cycles, we implement an optimization algorithm. We choose steepest descent, as it is suitable for our system for reasons discussed in Section Optimization Methods. After implementing the method, we test the algorithm by checking wether it can find the known optimal variational parameter.

Since consecutive samples in a Monte Carlo chain are correlated, we cannot apply the standard equations for the variance and standard deviation to obtain a reliable uncertainty. This is because these equations do not account for the correlation between samples and therefore the variance will be lower than the true value. To obtain a more accurate estimate of the uncertainty in the calculated energy, we implement a blocking method. Details about this method is in Section Statistical analysis with the blocking method.

Finally we implement interactions between the atoms in an elliptical trap. After determining the optimal variational parameter for the case with interactions and for $N \in \{10, 50, 100\}$ atoms, we investigate how the interactions affect the average spatial distribution of the atoms. To do this we simulate the trapped gas for $N \in \{10, 50, 100\}$ atoms, with and without interactions, and calculate and plot the column densities for both cases. We then compare our results with [6]. The details of column densities are given in Section Column Density.

Use when talking about: [4] VMC for BECs variational wavefunctions comparison with mean-field

Use [6] Nilsen (2005) Use when discussing: interaction effects beyond weakly interacting regime and role of correlations (Jastrow relevance!)

Results

All the results were obtained with the code available on GitHub repository

Brute Force Metropolis Results

Energy as a Function of the Variational Parameter α

Figure 1 shows the expectation value of the energy as a function of the variational parameter α for the one–dimensional harmonic oscillator with $N = 100$ atoms. The minimum of this curve is at $\alpha = 0.5$ and corresponds to the optimal variational parameter.

VMC vs exact values

In Table 1, the expectation values of the energy obtained using VMC are compared with the exact energies. For all $d \in \{1, 2, 3\}$ and $N \in \{1, 10, 100, 500\}$, the VMC energies match the exact energies to numerical precision.

d	N	E_{VMC}	E_{exact}	$\lvert E_{\text{VMC}} - E_{\text{exact}}\rvert$
1	1	0.500	0.500	0.000
1	10	5.000	5.000	0.000
1	100	50.000	50.000	0.000
1	500	250.000	250.000	0.000
2	1	1.000	1.000	0.000
2	10	10.000	10.000	0.000
2	100	100.000	100.000	0.000
2	500	500.000	500.000	0.000
3	1	1.500	1.500	0.000
3	10	15.000	15.000	0.000
3	100	150.000	150.000	0.000
3	500	750.000	750.000	0.000

Table 1: Comparison between VMC energies (E_{VMC}) and exact energies (E_{exact}), for non-interacting bosons in a spherical harmonic oscillator potential. N is number of particles and d is the number of dimensions for the given simulation. We used $100,000$ MC cycles for each configuration, (N,d). For the exact energies we have used Equation (13)

.

CPU Time Comparison

The analytical and numerical computation times per MC cycle for the energy are presented in Table 2. Since the potential energy is computed identically in the analytical and numerical cases, the difference in computation time is due to the calculation of the kinetic energy. In Table 2, we observe that the analytical computation time remains approximately constant as N increases, whereas the numerical computation time increases with N.

N	Analytical time (s)	Numerical time (s)
1	1.97×10^{-5}	2.24×10^{-4}
10	2.03×10^{-5}	6.77×10^{-4}
50	2.08×10^{-5}	2.71×10^{-3}
100	1.90×10^{-5}	4.11×10^{-3}

Table 2: Comparison of computational time per Monte Carlo cycle for analytical and numerical evaluation of the energy in one dimension. Here, N denotes the number of particles. Each simulation uses 1.0×10^5 MC cycles.

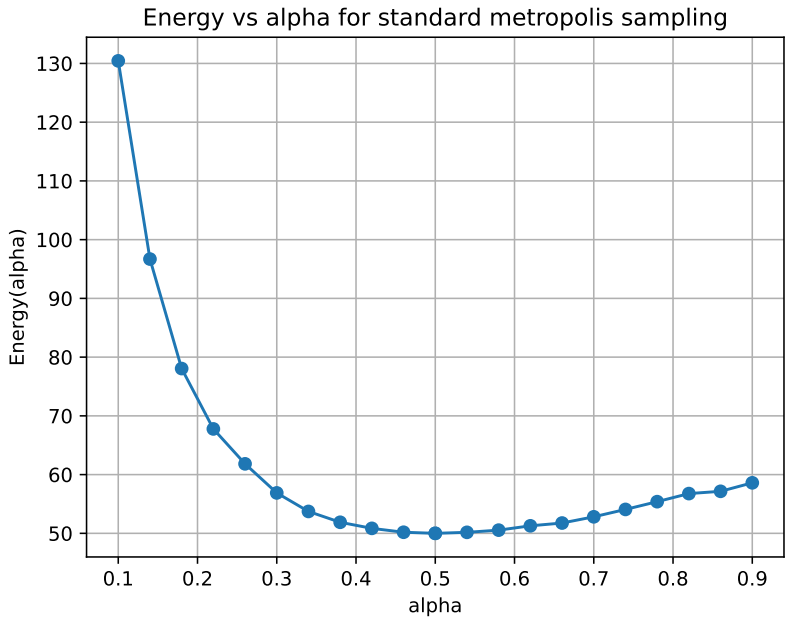

Figure 1: Expectation value of the energy as a function of the variational parameter α in one dimension for $N = 100$ particles using 5.0×10^5 MC cycles per alpha

Importance Sampling Results

Energy as a function of the variational parameter α (Importance sampling)

In Figure 2, a coarse scan of the variational parameter α for $N = 100$ is shown using importance sampling. The VMC energy is plotted as a function of α for 21 evenly spaced values of $\alpha \in [0.1, 0.9]$. The value $\alpha = 0.5$ with $E_{\text{VMC}} = 50.0$ yields the lowest VMC energy. All $\alpha \in [0.4, 0.6]$ have energy values close to the minimum.

Δt_{eff}	E_{VMC}	E_{exact}	Acceptance rate
0.0014	50.000	50.000	1.0000
0.0141	50.000	50.000	1.0000
0.1414	50.000	50.000	0.9790
0.7071	50.000	50.000	0.8650
1.4142	50.000	50.000	0.6310
2.8284	50.000	50.000	0.3400

Table 3: Dependence on the effective timestep $\Delta t_{\text{eff}} = \sqrt{1/\alpha}\,\Delta t$, for importance sampling in one dimension with $N = 100$ and $\alpha = 0.5$. For each Δt, 1.0×10^4 MC samples are used.

The Δt dependence of importance sampling

From Tables 3, 4, and 5 we observe that small Δt_{eff} values yield a 100% acceptance rate, and that the acceptance rate decreases as Δt_{eff} increases for one, two, and three dimensions. In one dimension (Table 3), the acceptance rate decreases from 100% to 34% as Δt_{eff} increases from 0.0014 to 2.8284, while in two (Table 4) and three (Table 5) dimensions, the acceptance rate decreases to 13.06% and 5.17%, respectively.

Δt_{eff}	E_{VMC}	E_{exact}	Acceptance rate
0.0014	100.000	100.000	1.0000
0.0141	100.000	100.000	0.9998
0.1414	100.000	100.000	0.9821
0.7071	100.000	100.000	0.7931
1.4142	100.000	100.000	0.4821
2.8284	100.000	100.000	0.1306

Table 4: Dependence on the effective timestep $\Delta t_{\text{eff}} = \sqrt{1/\alpha}\,\Delta t$ for importance sampling in two dimensions with $N = 100$ and $\alpha = 0.5$. Simulation was run with 1.0×10^4 samples for each Δt.

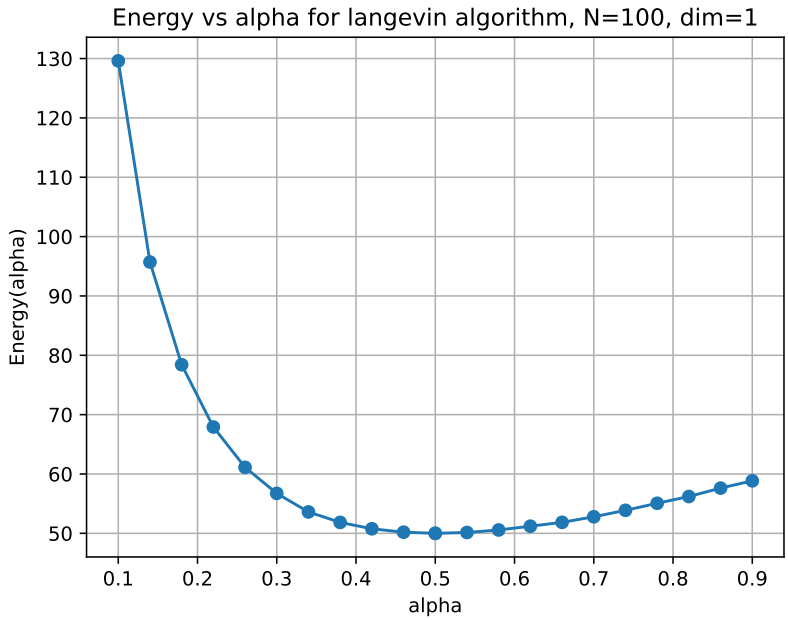

Figure 2: Expectation value of the energy as a function of the variational parameter α in one dimension for $N = 100$ particles. We used 5.0×10^5 MC cycles per alpha and the Metropolis Hastings algorithm for the sampling.

Δt_{eff}	E_{VMC}	E_{exact}	Acceptance rate
0.0014	150.000	150.000	1.0000
0.0141	150.000	150.000	0.9994
0.1414	150.000	150.000	0.9761
0.7071	150.000	150.000	0.7393
1.4142	150.000	150.000	0.3671
2.8284	150.000	150.000	0.0517

Table 5: Dependence on the effective timestep $\Delta t_{\text{eff}} = \sqrt{1/\alpha}\,\Delta t$ for importance sampling in three dimensions with $N = 100$ and $\alpha = 0.5$. Simulation was run with $10,000$ samples for each Δt.

N	$\langle E_{\text{VMC}} \rangle$	σ_E
10	15.000	$\sim 10^{-16}$
50	75.000	$\sim 10^{-16}$
100	150.000	$\sim 10^{-16}$

Table 6: Estimated VMC energy and statistical uncertainties from blocking analysis for the three-dimensional non-interacting spherical trap for $N \in \{10, 50, 100\}$ particles and $\alpha = 0.5$.

Optimization of Variational Parameters

Figure 3 presents alpha vs iterations, using steepest descent for $N = 100$ and with a spherical potential. The function converges to $\alpha = 0.5$ which we know as the optimal variational parameter.

Statistical Error Analysis

Figure 4 presents how the SEM of a spherical system with $N = 100$ particles and $\alpha = 0.5$, changes with block size B. We observe that the SEM is $< 10^{-18}$ which is expected since we have the exact ground state wavefunction for the harmonic oscillator. We Using the same argument $N = \{10, 50\}$ we obtain the results shown in Table 6, which presents the VMC energy (E_{VMC}), and its uncertainty σ_E for the spherical system with $N \in \{10, 50, 100\}$ for $\alpha = 0.5$.

Simulation of BEC with Cylindrical potential

In all the results below, we use $\beta = \gamma = \omega_z = 2.82843$ for the elliptical trap.

Comparing the energies in Table 7, we observe that the energies in the interacting case are higher then in the non interacting case for all N. We also observe that, in the interacting case, the optimal variational parameter α_{opt} is decreasing while the SEM is increasing, for increasing N. In the non-interacting case, both of these quantities remain constant for all N. The energy per particle is also constant for the non-interacting case whereas it increases with N in the interacting case.

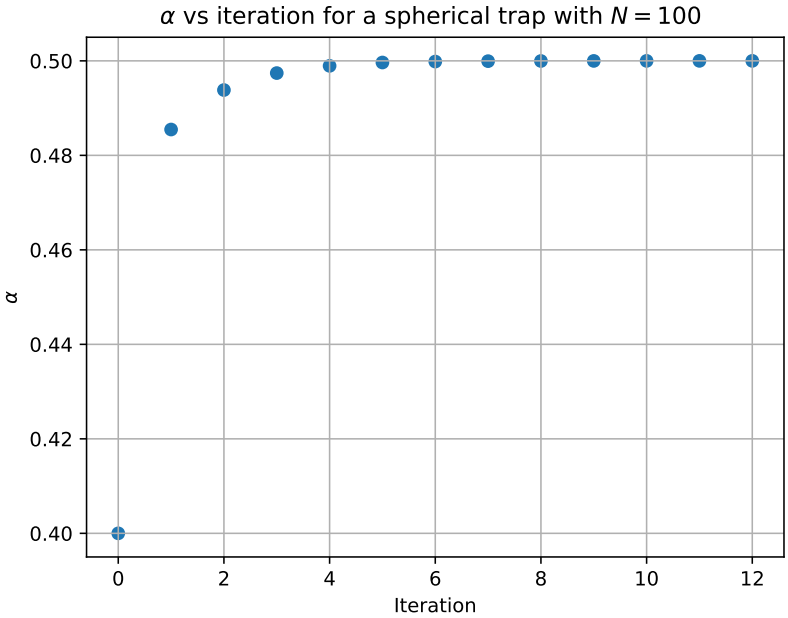

Figure 3: Plot of the variational parameter α over iterations for a spherical trap with $N = 100$ particles, using the steepest descent optimizer.

N	a	α_{opt}	E_{VMC}	SEM
10	0	0.500	24.142	$\sim 10^{-16}$
50	0	0.500	120.711	$\sim 10^{-16}$
100	0	0.500	241.422	$\sim 10^{-16}$
10	0.0043	0.497	24.405	1.09×10^{-3}
50	0.0043	0.489	127.260	6.54×10^{-3}
100	0.0043	0.483	266.17	1.95×10^{-2}

Table 7: Optimal variational parameters, energies, and statistical uncertainties for the elliptical trap with and without interactions. A learning rate of $\eta = 0.0075$ was used for $N = 10$, while $\eta = 0.001$ was used for the other simulations. All simulations use 1.0×10^6 MC cycles.

Without particle interaction

Using steepest descent we found that the optimal variational parameter with $a = 0.0$ is $\alpha = 0.5$ for particle numbers $N = \{10, 50, 100\}$. A representative convergence plot can be seen in Figure 5 showing that the system converges to $\alpha = 0.5$ for $N = 100$.

Figures 6, 7, and 8 show the column density for $N \in \{10, 50, 100\}$ particles, respectively. These plots correspond to a system without particle interactions. The column density is nearly identical for all numbers N. The distribution resembles a Gaussian with maximum at $r_{\text{perp}} = 0$, approaching zero at $r_{\text{perp}} \approx 2.5$. The only notable difference between the three plots is that the maximum value of n_c increases slightly with the number N.

With particle interactions

To determine the optimal variational parameter α for the interacting case, we used the steepest descent method with $\alpha_0 = 0.4$. Figure 9 shows that the system converges for $\alpha \approx 0.48$, with $N = 100$. Similar convergence behavior is observed for $N \in \{10, 50\}$ with slightly different optimal values of α.

The blocking analysis for $a = 0.0043$ with $N = 100$ is shown in Figure 10. We observe that the estimated SEM increases with block size until it converges in the region $B \approx 2000$–4000. Since the SEM is approximately constant in this region we take the value at the plateau as the best estimate of the SEM. We use the value at $B \approx 4000$ since the value at $B \approx 8000$ is affected by statistical noise. This yields $\sigma_{N=100} \approx 1.95 \times 10^{-2}$. Using the same procedure for $N \in \{10, 50\}$ yields $\sigma_{N=10} \approx 1.09 \times 10^{-3}$ and $\sigma_{N=50} \approx 6.54 \times 10^{-3}$.

Figures 11, 12 and 13 show the column density for $N = \{10, 50, 100\}$ particles, respectively. These plots correspond to a system with particle interactions. In all three plots the shape of the distribution is approximately Gaussian with a maximum for $r_{\text{perp}} = 0$ and approaches zero around $r_{\text{perp}} \approx 2.5$. The maximum value decreases with increasing N: for $N = 10$, $n_c \approx 0.310$, for $N = 50$, $n_c \approx 0.275$, and for $N = 100$, $n_c \approx 0.260$. Since all distributions approach zero around $r_{\text{perp}} \approx 2.5$ while having different maximum values, the distributions become broader as N increases.

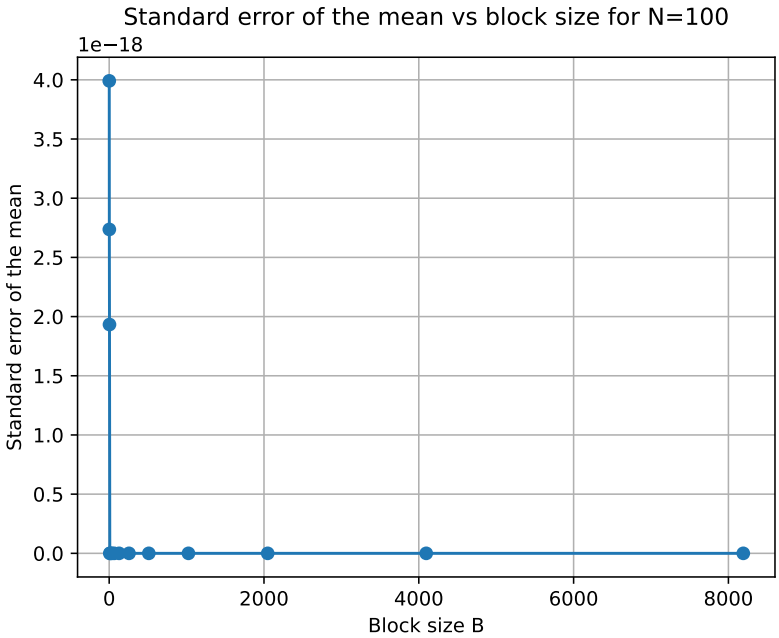

Figure 4: Plot of standard error of the mean (SEM) of a spherical system with $N = 100$ particles vs block size B using the variational parameter $\alpha = 0.5$ and 1.0×10^6 MC cycles.

Discussion

For the spherical non-interacting case the variational parameter is known exactly and the since we use the analytical expression for the energy the E_{VMC} is equal to the E_{exact} for all dimensions and number of particles. When comparing the analytical and numerical computation time for the energy the analytical time did not change for increasing N. This could be due to the code being vectorized and that all particle contributions to be evaluated simultaneously. In contrast, the computational time for the numerical case increases by a factor of approximately 20 from $N = 1$ to $N = 100$ particles. This is likely due to the numerical calculations involving gradients and Hessians, which scales poorly with increased N.

One possible reason why the acceptance rate decreases as you increase dimensions for importance sampling, is that the configuration space increases in size. As a result, a larger portion of the proposed moves will have a less probable configuration and therefore fewer moves is accepted in total [5].

When estimating the error of the energy E_{VMC} in the non interacting case the blocking method yields a $SEM = 0$ for both the spherical and elliptical system. This is because our chosen trial wavefunction is the exact ground state of the system and this yields a constant local energy. A constant local energy yields a $SEM = 0$, which means that our only error is the systematic error from the numerical precision and that the effective SEM is of order 10^{-16}.

One reason why the energies are overall higher in the interacting case compared to the non-interacting case is the presence of repulsive interactions between the atoms. From Equation (34), we see that to minimize the energy the system will increase the average distance between the atoms. This will lead to the atoms being further out from the center of the trap leading to an increase in the potential energy and therefore an overall increase in the total energy. This could also be the reason why the optimal parameter α_{opt} decreases with increasing N. As N increases, the effect of repulsive interactions becomes stronger. Since the trial wavefunction $\Psi_T(R) \sim e^{-\alpha_{opt}R^2}$ determines the probability distribution of the positions through the Born rule, a decrease in α_{opt} will lead to an increase in the average distance between the atoms.

The increase in SEM with increasing N, in the interacting case, is likely due to large fluctuations in the local energy caused by particle interactions. As N increase, the number of interacting particle pairs will lead to higher fluctuations in the local energy, resulting in an increase in the SEM.

Comparing the column densities for the interacting and non-interacting cases we observe that the distributions in the interacting systems are broader than in the non-interacting system. The discrepancy becomes more apparent with increasing interaction strength which is visible when comparing for $N = 10$ 11, with $N = 100$ 13. Here, we observe that the maximum is at approximately $n_c \approx 0.31$ for $N = 10$ and approximately $n_c \approx 0.26$ for $N = 100$. We therefore choose to focus on the $N = 100$ case. The distribution for the interacting case in Figure 13 is wider compared to the non-interacting case in Figure 8. This means that the particles are on average further away from the center of the trap which validates the claim made above for the decrease in α_{opt}. Looking at where the distributions goes to zero, we observe that in the interacting case the distribution goes to zero at a higher value

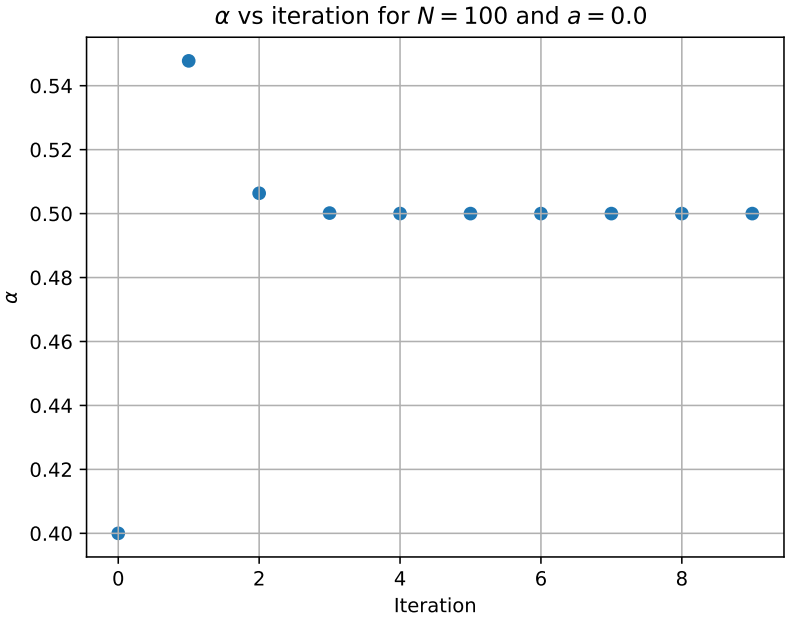

Figure 5: Alpha vs iteration using steepest descent and 1.0×10^5 MC cycles per alpha. Learning rate used is $\eta = 0.001$

of r_{perp} then in the non-interacting case. This means that the BEC with interactions are slightly larger then the BEC without interactions.

Comparing our energy per particle for the interacting case with the energy per particle in [4], we see that they also have an increase in this quantity for increasing N, which confirms that this is not a error from our simulation. Comparing our results with the results from [6], we observe that our result matches theirs. They simulated the BEC with interactions for $N = 500$ and got that the optimal parameter was $\alpha_{opt} = 0.475$. This matches our result since it is well below our α_{opt} for $N = 100$ and we also had a decrease in α_{opt} for an increase in N. When looking the column density plot on page 8 in [6], we observe that for $N = 500$ particles, the BEC has slightly grown in size compared to the non-interacting case. This matches our results for the case with interactions and $N = 100$.

Conclusion

To conclude, we have implemented and tested optimization tools and two different sampling algorithms to simulate BECs. We have compared steepest descent to coarse scan and provided evidence that you need less computational effort for finding the optimal variational parameter for our trial wavefunction with steepest descent. Furthermore, we have tested numerical and analytical computation times and found that for $N = 100$ the numerical computation times are 2×10^2 longer than the analytical. In addition, we investigated the dependence of Δt in importance sampling and discovered that the acceptance rate for higher dimensions decreases for large Δt. Importance sampling generally improves efficiency by guiding the sampling towards high-probability regions compared to brute-force Metropolis, which is why we used it for most of our simulations. Moreover, we simulated a BEC with and without interactions for multiple numbers of atoms and made plots of the respective column densities. With the use of the optimal variational parameter and the mathematical structure of our trial wavefunction and local energy, we made statements of the behaviour of the simulated BEC. With comparisons to scientific articles and plots of the columns densities we have confirmed that our simulations are correct. The change in the optimal variational parameter and the plots of the column densities provided evidence that the correlations between interacting particles, indeed have consequences for the BEC. We found that it not only increases the average distance between particles, but also slightly increases the size of the BEC for just $N = 100$ particles. We simulated the BEC with $N \in \{10, 50, 100\}$ particles and saw a clear correlation between increasing N, interaction effects and the average distance between the atoms. This leads us to conclude that for a BEC with many thousands of atoms, interactions would significantly increase the total volume of the condensate. This change in the physical properties of the BEC further motivates the use of VMC simulations when experimenting with tuning of interaction strength.

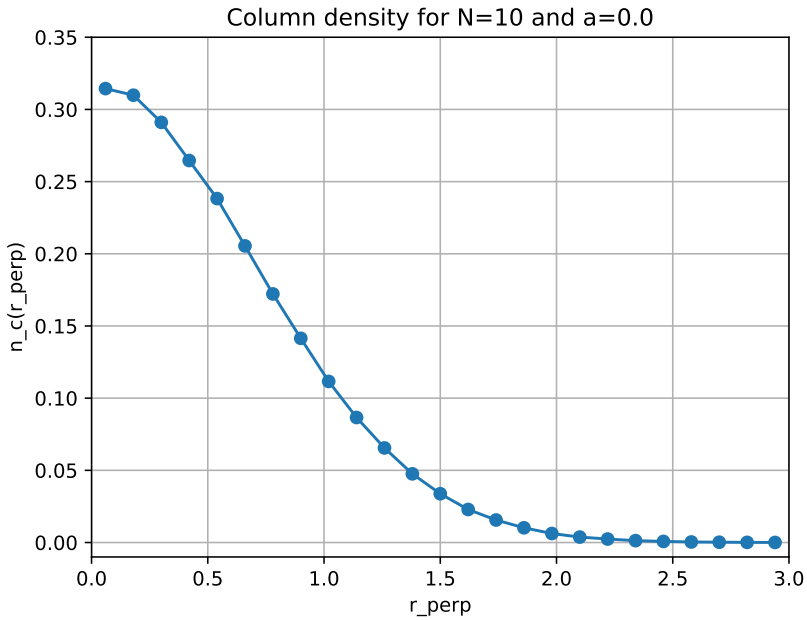

Figure 6: Column density n_c, without particle interactions for $N = 10$ particles. $r_{\text{perp}} = \sqrt{x^2 + y^2}$ represents the radial distance from the center of the trap in the x, y-plane given in harmonic oscillator units a_{ho}.

References

[1] I. Georgescu, *25 years of bose–einstein condensation*, *Nature Reviews Physics* **2** (2020) 396.

[2] J. E. Debs, P. A. Altin, *et. al.*, *Cold-atom gravimetry with a bose-einstein condensate*, *Physical Review A* **84** (2011).

[3] Computational Physics 2, "Project 1: Variational monte carlo for bosonic systems." `https://github.com/CompPhysics/ComputationalPhysics2/blob/gh-pages/doc/Projects/2026/Project1/ipynb/Project1.ipynb`, 2026. Accessed: 2026-03-25.

[4] J. L. DuBois and H. R. Glyde, *Bose-einstein condensation in trapped bosons: A variational monte carlo analysis*, *Physical Review A* **63** (2001) 023602.

[5] M. Hjorth-Jensen, *Computational physics 2: Lecture notes*, 2021. University lecture notes, accessed 2026-03-19.

[6] J. K. Nilsen, J. Mur-Petit, M. Guilleumas, M. Hjorth-Jensen, and A. Polls, *Vortices in atomic bose-einstein condensates in the large-gas-parameter region*, *Physical Review A* **71** (2005) 053610.

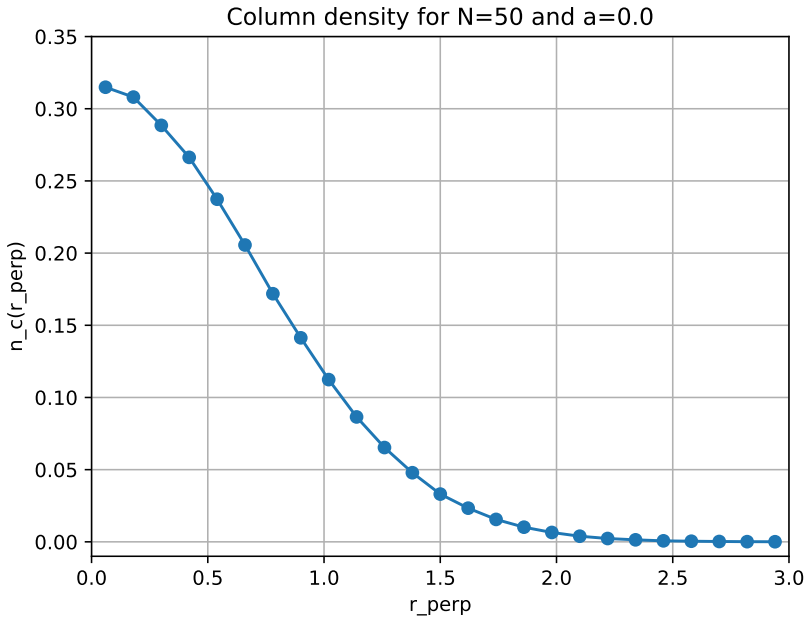

Figure 7: Column density n_c, without particle interactions for $N = 50$ particles. $r_{\text{perp}} = \sqrt{x^2 + y^2}$ represents the radial distance from the center of the trap in the x, y-plane given in harmonic oscillator units a_{ho}..

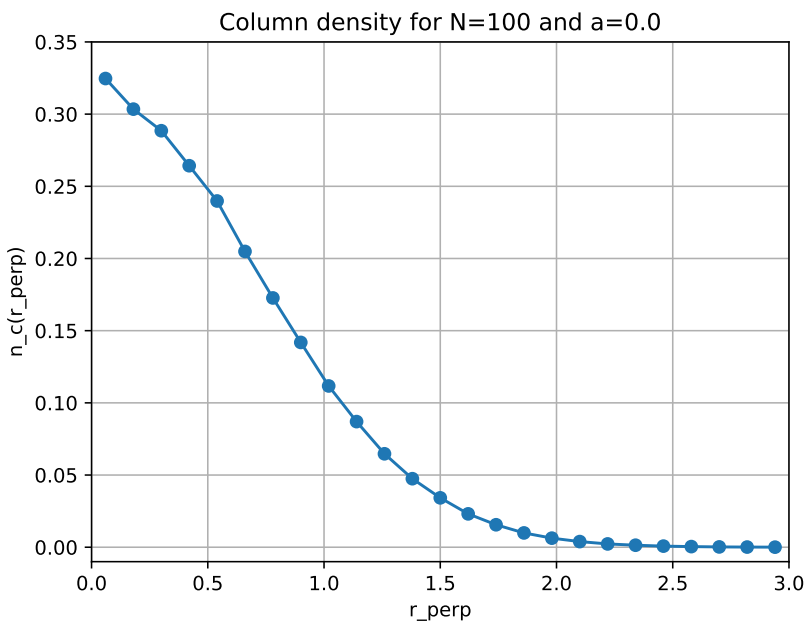

Figure 8: Column density n_c, without particle interactions for $N = 100$ particles. $r_{\text{perp}} = \sqrt{x^2 + y^2}$ represents the radial distance from the center of the trap in the x, y-plane given in harmonic oscillator units a_{ho}..

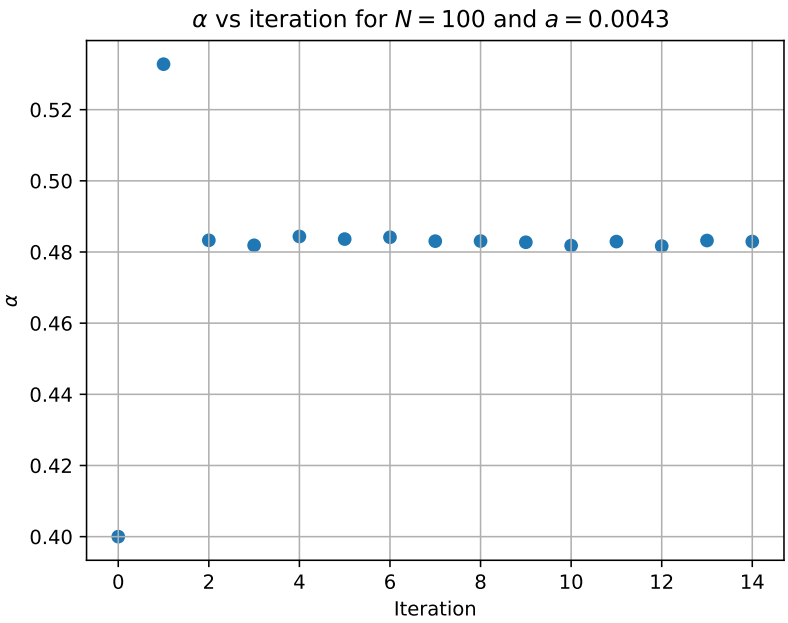

Figure 9: Alpha vs iteration using steepest descent and 1.0×10^5 MC cycles per alpha. Learning rate used is $\eta = 0.001$

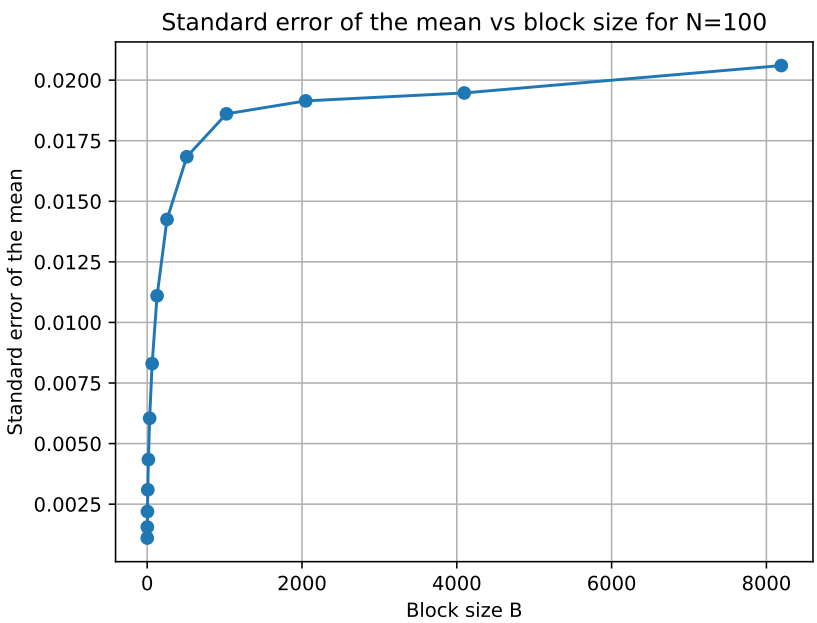

Figure 10: Standard error of the mean (SEM) as a function of block size (B) for, $N = 100$ particles, an elliptical trap and $a = 0.0043$. The simulation use 1.0×10^6 MC cycles

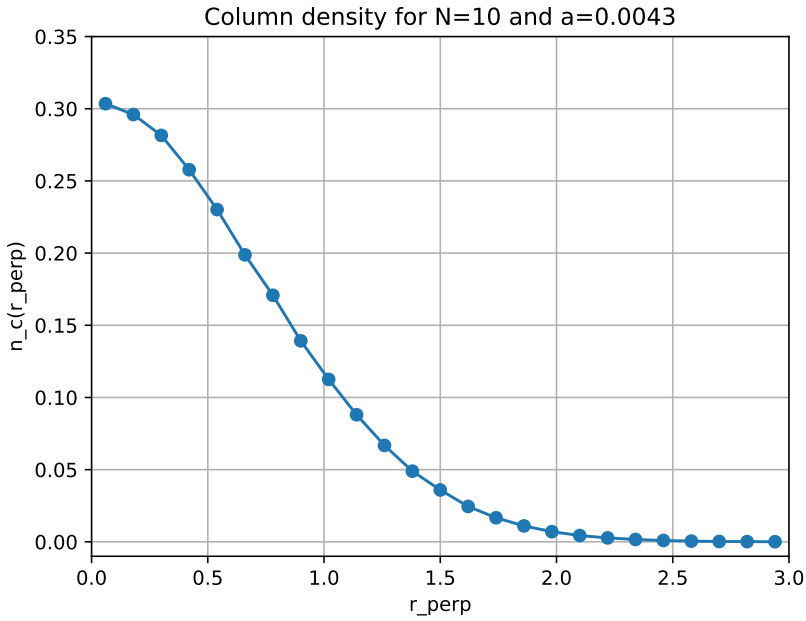

Figure 11: Column density n_c, with particle interactions for $N = 10$ particles. $r_{\text{perp}} = \sqrt{x^2 + y^2}$ represents the radial distance from the center of the trap in the x, y-plane given in harmonic oscillator units a_{ho}.

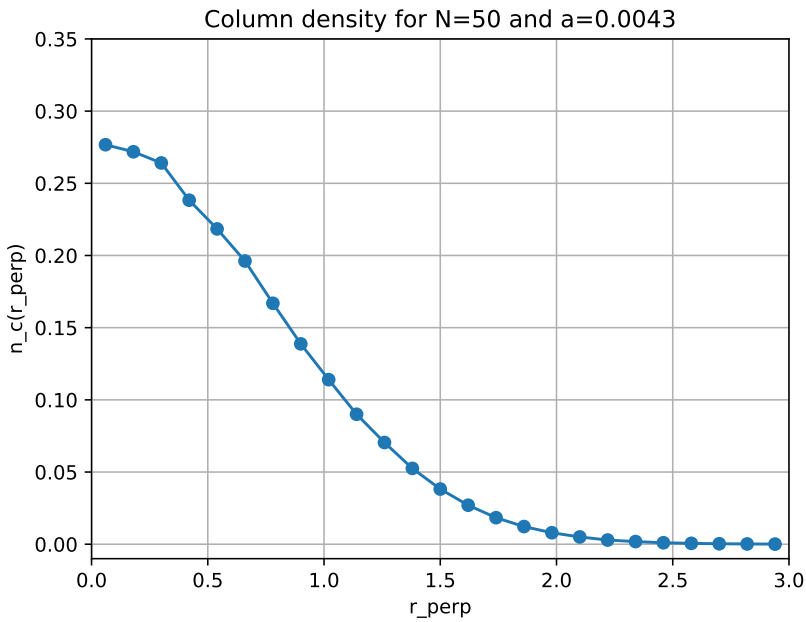

Figure 12: Column density n_c, with particle interactions for $N = 50$ particles. $r_{\text{perp}} = \sqrt{x^2 + y^2}$ represents the radial distance from the center of the trap in the x, y-plane given in harmonic oscillator units a_{ho}.

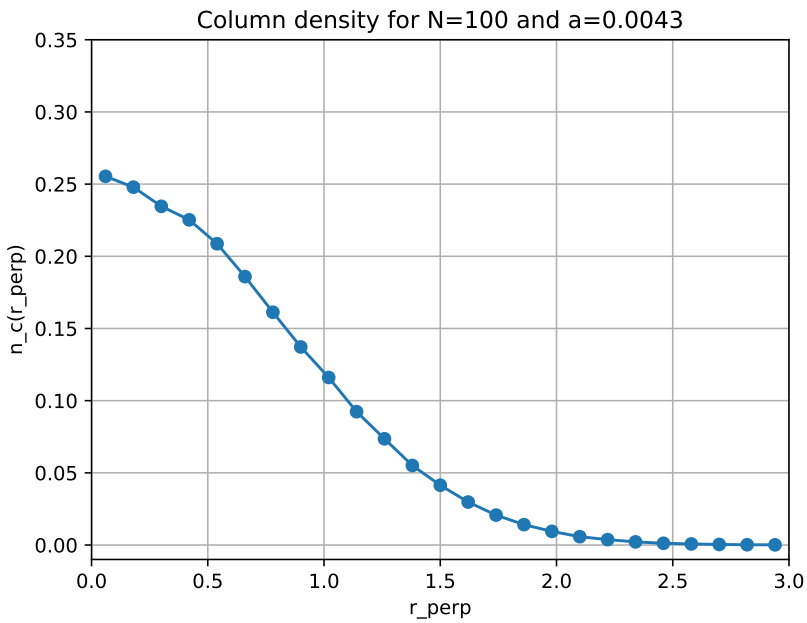

Figure 13: Column density n_c, with particle interactions for $N = 100$ particles. $r_{\text{perp}} = \sqrt{x^2 + y^2}$ represents the radial distance from the center of the trap in the x, y-plane given in harmonic oscillator units a_{ho}.

Appendix

Local energy with and without interactions

Local energy on logform is

$$E_L = -\frac{1}{2}\sum_i \left[\nabla_i^2 \ln \Psi_T + (\nabla_i \ln \Psi_T)^2\right] + \sum_i V_{\text{ext}}(r_i) + \sum_{i<j} V_{\text{int}}(r_{ij}) \tag{32}$$

Without interactions $V_{\text{int}} = 0$ and the wavefunction reduces to Equation(9) which leads to the following local energy for N particles without interactions:

$$E_L = \sum_{i=1}^{N} \alpha(2+\beta) - 2\alpha^2(x_i^2 + y_i^2 + \beta^2 z_i^2) + \frac{1}{2}\omega_{ho}^2(x_i^2 + y_i^2) + \frac{1}{2}\omega_z^2 z_i^2 \tag{33}$$

where the sum goes over all particles in the system.

The full expression for the local energy with interactions is

$$E_L(\mathbf{R}) = \sum_{k=1}^{N} \left[\alpha(2+\beta) - 2\alpha^2(x_k^2 + y_k^2 + \beta^2 z_k^2)\right] \tag{34}$$

$$+ \frac{1}{2}\omega_{ho}^2 \sum_{k=1}^{N}(x_k^2 + y_k^2) + \frac{1}{2}\omega_z^2 \sum_{k=1}^{N} z_k^2 \tag{35}$$

$$+ 2\alpha \sum_{k=1}^{N}(x_k, y_k, \beta z_k) \cdot \sum_{j\neq k} \frac{a(\mathbf{r}_k - \mathbf{r}_j)}{r_{kj}^2(r_{kj} - a)} \tag{36}$$

$$- \frac{1}{2}\sum_{k=1}^{N} \sum_{\substack{i\neq k \\ j\neq k \\ i\neq j}} \frac{a^2(\mathbf{r}_k - \mathbf{r}_i)\cdot(\mathbf{r}_k - \mathbf{r}_j)}{r_{ki}^2 r_{kj}^2 (r_{ki} - a)(r_{kj} - a)}. \tag{37}$$

Quantum force for Langevin drift

Quantum force without interactions is

$$\mathbf{F}_i = -4\alpha\,(x_i,\, y_i,\, \beta z_i). \tag{38}$$

Quantum force with interactions is

$$\mathbf{F}_i^{int} = -4\alpha\,(x_i,\, y_i,\, \beta z_i) + 2\sum_{j\neq i} \frac{a}{r_{ij}^2(r_{ij} - a)}(\mathbf{r}_i - \mathbf{r}_j), \tag{39}$$